

BARMER, IndiaWMO: **424350**Lat: **25.750N**Lon: **71.383E**Elev: **194**StdP: **99.02**Time Zone: **5.50 (IND)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.1	11.3	-5.6	2.4	18.0	-3.3	2.9	18.2	2.8	21.2	2.4	20.8	0.1	N/A	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	12.3	43.2	23.3	41.9	23.2	40.6	23.4	28.0	34.5	27.6	34.1	27.2	33.8	1.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	22.9	30.3	26.2	22.2	30.2	25.7	21.5	30.2	91.7	34.6	89.5	34.0	87.7	33.7	30.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 3.4	2.9	2.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			2.8	29.2	1.7	0.8	1.6	29.8	0.6	30.3	-0.3	30.7	-1.6	31.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.0	17.9	21.7	27.9	32.9	35.0	34.6	32.6	30.5	30.7	29.1	24.0	19.1
	DBStd	6.16	2.33	3.05	3.27	2.42	2.10	2.02	2.03	1.49	1.76	2.01	2.44	2.49
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	61	37	4	0	0	0	0	0	0	0	0	0	19
	CDD10.0	6582	246	327	556	687	777	738	699	637	622	591	419	283
	CDD18.3	3601	24	98	298	437	518	488	441	378	372	333	169	44
	CDH23.3	52889	155	882	3899	7099	8862	8108	6844	5318	5375	4478	1589	280
	CDH26.7	32742	14	287	2169	4792	6404	5719	4382	2874	3039	2431	576	56
Wind	WSAvg	0.5	0.3	0.4	0.4	0.7	1.1	1.0	1.0	0.6	0.5	0.2	0.1	0.1
Precipitation	PrecAvg	300	1	4	2	5	9	29	83	90	53	7	3	1
	PrecMax	733	6	33	17	68	71	122	289	497	216	49	65	16
	PrecMin	48	0	0	0	0	0	0	0	0	1	0	0	0
	PrecStd	156	2	7	4	13	17	35	77	108	63	13	13	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.4	34.3	39.9	43.5	45.2	45.1	41.6	37.2	38.6	37.6	34.1	30.4
		MCWB	17.1	18.4	21.1	21.8	23.0	23.9	26.3	26.4	25.2	22.7	19.5	18.1
	2%	DB	26.5	32.0	38.2	41.9	43.3	43.0	39.4	36.1	37.5	36.5	32.4	28.2
		MCWB	15.5	17.9	20.4	21.5	23.0	25.2	26.6	26.2	24.9	22.2	19.2	17.2
	5%	DB	25.2	30.1	36.7	40.7	42.2	41.2	38.1	35.2	36.4	35.4	31.1	26.3
		MCWB	14.8	17.2	19.5	21.3	22.7	25.4	26.3	26.0	25.0	21.8	18.8	16.2
	10%	DB	23.7	28.2	35.0	39.3	40.9	39.6	37.0	34.3	35.2	34.5	29.8	24.5
		MCWB	14.0	16.3	19.0	21.4	23.1	25.6	26.1	26.0	24.9	21.4	17.9	15.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.7	20.8	23.7	25.7	27.5	28.6	28.5	28.4	28.2	26.7	23.4	20.2
		MCDB	25.9	30.4	33.3	38.2	38.0	37.1	34.4	32.7	33.2	32.5	28.6	27.2
	2%	WB	17.0	19.8	22.1	24.6	26.7	27.9	27.9	27.7	27.4	25.9	22.2	18.3
		MCDB	24.3	28.9	34.4	36.9	37.2	36.5	34.2	32.4	32.6	31.4	28.4	25.5
	5%	WB	15.9	18.6	21.0	23.7	26.1	27.3	27.5	27.2	26.8	24.7	20.7	17.2
		MCDB	23.2	27.2	33.9	35.8	36.2	36.1	34.0	31.8	32.0	31.3	28.2	24.9
	10%	WB	14.8	17.5	20.1	22.9	25.6	26.9	27.1	26.9	26.3	23.8	19.5	16.0
		MCDB	22.4	25.9	32.8	35.1	35.7	35.8	33.8	31.5	31.6	31.4	28.0	23.6
Mean Daily Temperature Range	5% DB	MDBR	11.6	12.1	12.4	12.0	12.3	10.3	8.3	6.7	8.1	10.2	10.5	11.1
		MCDBR	12.8	13.6	13.2	13.1	13.2	12.3	9.9	8.3	9.8	10.9	11.9	12.7
		MCWBR	6.1	5.8	5.0	4.3	3.7	3.1	1.8	2.0	2.6	4.1	5.2	6.5
		MCWBR	11.6	11.2	12.1	12.1	12.6	10.5	8.1	6.8	7.6	9.1	9.6	12.0
	5% WB	MCWBR	6.3	5.5	5.3	4.9	3.8	2.9	2.4	2.5	2.5	4.1	4.8	6.8
Clear-Sky Solar Irradiance	taub	0.407	0.434	0.489	0.560	0.620	0.693	0.735	0.656	0.566	0.507	0.478	0.430	
	taud	2.070	2.019	1.885	1.761	1.681	1.600	1.561	1.708	1.864	1.885	1.905	2.008	
	Ebn at Noon	826	835	810	763	717	661	633	683	736	754	743	780	
	Edh at Noon	145	164	198	229	248	267	277	239	199	185	170	148	
All-Sky Solar Radiation	RadAvg	4.28	5.17	6.23	6.89	7.13	6.51	5.37	5.34	5.72	5.51	4.45	4.04	
	RadStd	0.16	0.18	0.18	0.19	0.18	0.30	0.35	0.33	0.31	0.17	0.23	0.16	